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ANTENNA DEVICE

Abstract

An antenna device includes a circuit board, a coil antenna and a resonance circuit. The circuit board has a conductive portion with a flat shape. The conductive portion is formed with an opening and a slit. The slit is coupled with the opening. The coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along an up-down direction perpendicular to the circuit board. The coil antenna includes a coil and a magnetic core. The magnetic core partially forms a magnetic path of the coil. The magnetic core is positioned at a region excluding a specific region which is positioned just below the coil. The magnetic core at least has a center core. The center core is at least partially positioned inside the coil.

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Background/Summary

TECHNICAL FIELD

[0001] This invention relates to an antenna device comprising a circuit board and a coil antenna.

BACKGROUND ART

[0002] As shown in FIG. 31, an antenna device 900 of this type is disclosed in Patent Document 1. The antenna device 900 can be used for communications such as NFC (Near field communication) or RFID (radio frequency identifier). The antenna device 900 comprises a circuit board 910, a first coil antenna (coil antenna) 930 and a second coil antenna (coil antenna) 940. As shown in FIG. 32, a first principal surface 920 of the circuit board 910 is formed with a planar conductor (conductive portion) 922. The conductive portion 922 is formed with a first opening (opening) 9222, a second opening (opening) 9224, a first slit (slit) 9223 and a second slit (slit) 9225. The slit 9223 is coupled with the opening 9222. The slit 9225 is coupled with the opening 9224. Referring to FIGS. 31 and 32, the coil antenna 930 is mounted on the circuit board 910 so that the coil antenna 930 overlaps with the opening 9222 when the coil antenna 930 is viewed along a direction perpendicular to the circuit board 910, or along a Z-direction. The coil antenna 940 is mounted on the circuit board 910 so that the coil antenna 940 overlaps with the opening 9224 when the coil antenna 940 is viewed along the direction perpendicular to the circuit board 910, or along the Z-direction. Each of the coil antennas 930, 940 comprises a coil (not shown) and a laminated body (not shown). Specifically, the coil has a winding axis extending in a direction perpendicular to the first principal surface 920, or in the Z-direction, and the laminated body is formed by laminating insulation base layers each made of magnetic ferrite or the like.

PRIOR ART DOCUMENTS

Patent Document(s)

[0003] Patent Document 1: JP-B 6566184

SUMMARY OF INVENTION

Technical Problem

[0004] An antenna device such as the antenna device 900 of Patent document 1 is required to have further enhanced antenna characteristics.

[0005] It is therefore an object of the present invention to provide an antenna device having further enhanced antenna characteristics.

Solution to Problem

[0006] In a coil antenna whose coil is completely embedded in a magnetic core, a closed magnetic path is formed by the magnetic core. The inventors of the present invention have noticed that, when an antenna device such as the antenna device 900 is formed by such a coil antenna, magnetic flux generated by a coil passes in a closed magnetic path formed by a magnetic core and this weakens coupling of a current-loop, which flows at an outer periphery of a slit of a circuit board, with the coil antenna. Then, the inventors of the present invention have conceived that an antenna device is configured to have an open magnetic path, through which magnetic flux generated by a coil of a coil antenna passes, by a magnetic core being at least not positioned just below the coil of the coil antenna. The inventors of the present invention have found that this new configuration can strengthen coupling of a current-loop, which flows at an outer periphery of a slit of a circuit board,

with the coil antenna. The present invention is based on this finding.

[0007] An aspect of the present invention provides, as a first antenna device, an antenna device comprising a circuit board, a coil antenna and a resonance circuit, wherein: [0008] the circuit board has a conductive portion with a flat shape; [0009] the conductive portion is formed with an opening and a slit; [0010] the conductive portion has a first path which goes around the opening and the slit; [0011] the first path forms a part of the resonance circuit; [0012] the slit is coupled with the opening; [0013] the coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along an up-down direction perpendicular to the circuit board; [0014] the coil antenna comprises a coil and a magnetic core; [0015] the coil has a winding axis extending in the up-down direction; [0016] the magnetic core partially forms a magnetic path of the coil; [0017] the magnetic core is positioned at a region excluding a specific region which is positioned just below the coil; [0018] the magnetic core at least has a center core; and [0019] the center core is at least partially positioned inside the coil.

Advantageous Effects of Invention

[0020] The antenna device of the present invention is configured as follows: the coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along the up-down direction perpendicular to the circuit board; the magnetic core is positioned at the region excluding the specific region which is positioned just below the coil; the magnetic core at least has the center core; and the center core is at least partially positioned inside the coil. This strengthens coupling of a current-loop, which flows at an outer periphery of the slit of the circuit board, with the coil antenna in the antenna device of the present invention. In other words, the antenna device of the present invention has further enhanced antenna characteristics.

[0021] An appreciation of the objectives of the present invention and a more complete understanding of its structure may be had by studying the following description of the preferred embodiment and by referring to the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0022] FIG. 1 is a perspective view showing an antenna device according to a first embodiment of the present invention.

[0023] FIG. 2 is a top view showing the antenna device of FIG. 1.

[0024] FIG. 3 is a cross-sectional view showing a part of the antenna device of FIG. 2, taken along line A-A. In the figure, a molded resinous member, terminals and pads are omitted.

[0025] FIG. 4 is a perspective view showing a coil antenna which is included in the antenna device of FIG. 1.

[0026] FIG. 5 is a side view showing the coil antenna of FIG. 4.

[0027] FIG. 6 is a top view showing the coil antenna of FIG. 4.

[0028] FIG. 7 is a front view showing the coil antenna of FIG. 4.

[0029] FIG. 8 is a perspective view showing a state where the molded resinous member is removed from the coil antenna of FIG. 4.

[0030] FIG. 9 is a side view showing the coil antenna of FIG. 8.

[0031] FIG. 10 is a top view showing the coil antenna of FIG. 8.

[0032] FIG. 11 is a front view showing the coil antenna of FIG. 8.

[0033] FIG. 12 is a top view showing a circuit board which is included in the antenna device of FIG. 2.

[0034] FIG. 13 is a perspective view showing a first modification of the coil antenna of FIG. 4.

[0035] FIG. 14 is a side view showing the coil antenna of FIG. 13.

[0036] FIG. **15** is a top view showing the coil antenna of FIG. **13**.
[0037] FIG. **16** is a front view showing the coil antenna of FIG. **13**.
[0038] FIG. **17** is a perspective view showing a state where a molded resinous member is removed from the coil antenna of FIG. **13**.
[0039] FIG. **18** is a side view showing the coil antenna of FIG. **17**.
[0040] FIG. **19** is a top view showing the coil antenna of FIG. **17**.
[0041] FIG. **20** is a cross-sectional view showing the coil antenna of FIG. **19**, taken along line B-B.
[0042] FIG. **21** is a front view showing the coil antenna of FIG. **17**.
[0043] FIG. **22** is a perspective view showing a second modification of the coil antenna of FIG. **4**.
[0044] FIG. **23** is a side view showing the coil antenna of FIG. **22**.
[0045] FIG. **24** is a top view showing the coil antenna of FIG. **22**.
[0046] FIG. **25** is a cross-sectional view showing the coil antenna of FIG. **24**, taken along line C-C. In the figure, a molded resinous member is omitted.
[0047] FIG. **26** is a front view showing the coil antenna of FIG. **22**.
[0048] FIG. **27** is a top view showing an antenna device according to a second embodiment of the present invention.
[0049] FIG. **28** is a top view showing an antenna device according to a third embodiment of the present invention.
[0050] FIG. **29** is a top view showing an antenna device according to a fourth embodiment of the present invention.
[0051] FIG. **30** is a top view showing an antenna device according to a fifth embodiment of the present invention.
[0052] FIG. **31** is a top view showing an antenna device of Patent Document **1**.
[0053] FIG. **32** is a top view showing a circuit board which is included in the antenna device of FIG. **31**.

DESCRIPTION OF EMBODIMENTS

[0054] While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

First embodiment

[0055] **20** As shown in FIG. **1**, an antenna device **10** according to a first embodiment of the present invention comprises a circuit board **500**, a coil antenna **100** and a resonance circuit.

[0056] As shown in FIG. **12**, the circuit board **500** of the present embodiment has a conductive portion **510** with a flat shape. In more detail, the conductive portion **510** is a ground conductor. Specifically, the circuit board **500** is configured so that a conductive layer, which functions as the conductive portion **510**, is laminated on an insulating substrate such as a glass epoxy substrate.

[0057] As shown in FIG. **12**, the conductive portion **510** is formed with an opening **512** and a slit **514**. The conductive portion **510** has a first path which goes around the opening **512** and the slit **514**. The first path forms a part of the resonance circuit.

[0058] Referring to FIG. **12**, the opening **512** of the present embodiment is a part of the circuit board **500** which has no conductive pattern. The opening **512** has an outer edge which has a substantially circular shape when the circuit board **500** is viewed from above in an up-down direction. However, the present invention is not limited thereto, but the opening **512** may have an outer edge with a shape other than the substantially circular shape. In the present embodiment, the up-down direction is a Z-direction. Specifically, it is assumed that upward is a positive Z-direction while downward is a negative Z-direction.

[0059] As shown in FIG. **12**, similar to the opening **512**, the slit **514** of the present embodiment is a

part of the circuit board **500** which has no conductive pattern. The slit **514** is coupled with the opening **512**. The slit **514** extends from the opening **512**. In the present embodiment, the number of the slit **514** extending from the opening **512** is one. However, the present invention is not limited thereto, but a plurality of the slits **514** may extend from the opening **512**. The slit **514** does not communicate with the outside of the conductive portion **510** in a plane perpendicular to the up-down direction. The slit **514** has a predetermined portion PP. The slit **514** has a first portion **5142** and a second portion **5144**.

[0060] As shown in FIG. **12**, a width of the slit **514** is smaller than a diameter of the opening **512**. However, the present invention is not limited thereto, but the width of the slit **514** may be equal to the diameter of the opening **512**, or the width of the slit **514** may be greater than the diameter of the opening **512**. It is noted that an increase of the width of the slit **514** enlarges a current loop, which is formed around the slit **514**, so that a communication range of the antenna device **10** is increased.

[0061] As shown in FIG. **12**, the first portion **5142** extends rightward in a right-left direction from the opening **512**. In the present embodiment, the right-left direction is a Y-direction. Specifically, rightward is a positive Y-direction while leftward is a negative Y-direction.

[0062] As shown in FIG. **12**, the second portion **5144** extends from the first portion **5142** in a direction intersecting with the first portion **5142**. The second portion **5144** extends rearward in a front-rear direction from a right end of the first portion **5142**. In the present embodiment, the front-rear direction is an X-direction. Specifically, forward is a positive X-direction while rearward is a negative X-direction. The second portion **5144** functions as the predetermined portion PP.

[0063] As shown in FIG. **12**, the slit **514** further comprises a third portion **5146**.

[0064] As shown in FIG. **12**, the third portion **5146** extends from the second portion **5144** in a direction intersecting with the second portion **5144**. The third portion **5146** extends leftward in the right-left direction from a rear end of the second portion **5144**.

[0065] As shown in FIG. **12**, the circuit board **500** has an edge **520**.

[0066] As shown in FIG. **12**, the edge **520** defines an outer edge of the circuit board **500**.

Specifically, the edge **520** defines the outer edge of the circuit board **500** in a direction perpendicular to the up-down direction. The slit **514** extends from the opening **512** toward the edge **520**, and is bent so that it extends along the edge **520**, and is further bent so that it extends away from the edge **520**. The slit **514** does not reach the edge **520**. The first portion **5142** extends from the opening **512** toward the edge **520**. The predetermined portion PP extends along the edge **520** in the vicinity of the edge **520**. This configuration enables the circuit board **500** to have an increased region where electronic components can be arranged on the circuit board **500**, and the communication range of the antenna device **10** can be increased by enlarging the current loop, which is formed around the slit **514**, because this configuration can increase a length of the slit **514**.

[0067] As described above, the slit **514** extends from the opening **512** toward the edge **520**, and is bent so that it extends along the edge **520**, and is further bent so that it extends away from the edge **520**. Additionally, as described above, the slit **514** does not reach the edge **520** and does not communicate with the outside of the conductive portion **510** in the plane perpendicular to the up-down direction. However, the present invention is not limited thereto, but the slit **514** may extend toward the edge **520** and may reach the edge **520**. Specifically, the slit **514** may communicate with the outside of the conductive portion **510** in the plane perpendicular to the up-down direction.

[0068] Referring to FIGS. **2** and **12**, the coil antenna **100** is mounted on the circuit board **500** so that the coil antenna **100** at least overlaps with the opening **512** when the coil antenna **100** is viewed along the up-down direction perpendicular to the circuit board **500**. The coil antenna **100** is surface-mounted on the circuit board **500**. In other words, the coil antenna **100** is a surface mount antenna.

[0069] As shown in FIG. **3**, the coil antenna **100** comprises a coil **200** and a magnetic core **300**.

[0070] As shown in FIG. **3**, the coil **200** of the present embodiment has a winding axis AX which extends in the up-down direction. This coil **200** is formed by winding a jacketed flat electrical cable

edgewise. However, the present invention is not limited thereto, but the coil **200** may be formed by winding a jacketed round electrical cable, and a material of the conductive cable, a cross-section of the conductive cable and a manner of winding the conductive cable are not limited to the present embodiment. The coil **200** has a ring shape when viewed along the up-down direction. However, the present invention is not limited thereto, but the coil **200** may have a shape other than the ring shape. The coil **200** has an upper end **210** and a lower end **220** in the up-down direction. The coil **200** has an inner periphery **240** and an outer periphery **230**.

[0071] Referring to FIGS. **3** and **12**, $S1$ and $S2$ satisfy $1 \leq S2/S1 \leq 2$, where $S1$ is a cross-sectional area of the inner periphery **240** of the coil **200**, and $S2$ is an area of the opening **512**. This can maximize a coupling coefficient between the coil antenna **100** and the slit **514**, and thereby the coil antenna **100** can easily be electromagnetically coupled with a current (current loop) which goes around the opening **512** and the slit **514** of the conductive portion **510**.

[0072] Referring to FIG. **3**, the magnetic core **300** of the present embodiment is a cast core. More specifically, the magnetic core **300** is a composite magnet in which soft magnetic particles made of ferrous alloy, ferrite or the like are dispersed in a hardened binder. In other words, the magnetic core **300** is formed by hardening a slurry which includes the soft magnetic particles and the binder or the like. It is noted that a method of manufacturing the magnetic core **300** is not limited to the aforementioned method and that the magnetic core **300** may be formed by any manufacturing method. The magnetic core **300** partially forms a magnetic path of the coil **200**. The magnetic core **300** is positioned at a region excluding a specific region SR which is positioned just below the coil **200**. In other words, the magnetic core **300** has no portion which is positioned just below the coil **200**. It is noted that the specific region SR is a region which is hidden below the coil **200** when the coil **200** is viewed alone from above. The specific region SR is positioned below the lower end **220** of the coil **200** in the up-down direction. The specific region SR is positioned outside the inner periphery **240** of the coil **200** in a radial direction perpendicular to the winding axis AX. The specific region SR is positioned inside the outer periphery **230** of the coil **200** in the radial direction perpendicular to the winding axis AX. The magnetic core **300** at least has a center core **310**.

[0073] Referring to FIG. **3**, the center core **310** of the present embodiment has a cylindrical shape extending in the up-down direction. The center core **310** is at least partially positioned inside the coil **200**. A center axis of the cylindrical shape of the center core **310** is coincide with the winding axis AX of the coil **200**. A size of a cross-section, which is perpendicular to the up-down direction, of the center core **310** is smaller than the diameter of the opening **512**. A projected image of the center core **310**, which is projected onto the circuit board **500** in the up-down direction, is positioned within the opening **512**.

[0074] As described above, the coil antenna **100** is mounted on the circuit board **500** so that the coil antenna **100** at least overlaps with the opening **512** when the coil antenna **100** is viewed along the up-down direction perpendicular to the circuit board **500**, the magnetic core **300** is positioned at the region excluding the specific region SR which is positioned just below the coil **200**, the magnetic core **300** at least has the center core **310** and the center core **310** is at least partially positioned inside the coil **200**. Thus, in the antenna device **10** of the present embodiment, a part of a flux of the coil **200** flows to the opening **512** of the circuit board **500** and thereby the coil antenna **100** can easily be electromagnetically coupled with the current (current loop) which goes around the opening **512** and the slit **514** of the conductive portion **510**. In other words, this strengthens coupling of the current-loop, which flows at an outer periphery of the slit **514** of the circuit board **500**, with the coil antenna **100** in the antenna device **10** of the present embodiment, and thereby the antenna device **10** of the present embodiment has further enhanced antenna characteristics.

[0075] As shown in FIG. **3**, the center core **310** extends downward in the up-down direction beyond the lower end **220** of the coil **200**. Thus, in the antenna device **10** of the present embodiment, the part of the flux of the coil **200** further flows to the opening **512** of the circuit board **500** and thereby the coil antenna **100** can more easily be electromagnetically coupled with

the current (current loop) which goes around the opening 512 and the slit 514 of the conductive portion 510. In other words, this further strengthens the coupling of the current-loop, which flows at the outer periphery of the slit 514 (see FIG. 12) of the circuit board 500, with the coil antenna 100 in the antenna device 10 of the present embodiment, and thereby the antenna device 10 of the present embodiment has more further enhanced antenna characteristics.

[0076] As shown in FIG. 3, the center core 310 extends upward in the up-down direction beyond the upper end 210 of the coil 200.

[0077] As shown in FIG. 3, the magnetic core 300 further has an upper portion 320.

[0078] As shown in FIG. 3, the upper portion 320 extends from the center core 310 toward the outer periphery 230 of the coil 200. In the radial direction perpendicular to the center axis of the cylindrical shape of the center core 310, an outer end of the upper portion 320 is positioned outside the outer periphery 230 of the coil 200. The upper portion 320 overlaps with the coil 200 when the upper portion 320 is viewed in the up-down direction. This enables the flux of the coil 200, which flows through the center core 310 to an upper part of the magnetic core 300, to efficiently flow toward the outer periphery 230 of the coil 200.

[0079] As shown in FIG. 7, the coil antenna 100 further has two terminals 400.

[0080] Referring to FIG. 8, the terminals 400 of the present embodiment are attached to end portions, respectively, of the coil 200. However, the present invention is not limited thereto, but the terminal 400 may be a portion formed by elongating the end portion of the coil 200. Each of the terminals 400 defines a lower end of the coil antenna 100. The terminals 400 are connected to a communication circuit (not shown) via pads of the circuit board 500.

[0081] As shown in FIG. 4, the coil antenna 100 comprises a molded resinous member 700.

[0082] Referring to FIGS. 5 and 9, the molded resinous member 700 covers the coil 200. Referring to FIGS. 3, 5 and 7, the specific region SR is occupied by the molded resinous member 700. In other words, the molded resinous member 700 is positioned just below the coil 200.

[0083] As shown in FIG. 2, the antenna device 10 further has a capacitor 600 which forms a part of the resonant circuit.

[0084] Referring to FIG. 12, the capacitor 600 forms a current loop path together with the first path which goes around the opening 512 and the slit 514 of the conductive portion 510. In other words, the capacitor 600 forms the resonant circuit together with the first path which goes around the opening 512 and the slit 514 of the conductive portion 510. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device 10. The capacitor 600 is mounted on the circuit board 500 at its region between the edge 520 and the slit 514. In more detail, the capacitor 600 is mounted on the circuit board 500 at its region between the edge 520 and the second portion 5144. However, the mounting location of the capacitor 600 is not limited to that described above, but the capacitor 600 may be mounted at any location on the conductive portion 510, provided that the capacitor 600 and the first path, which goes around the opening 512 and the slit 514 of the conductive portion 510, form the current loop path.

[0085] The configuration of the coil antenna 100 is not limited thereto, but the coil antenna 100 can be modified, for example, as described below.

First Modification

[0086] Referring to FIGS. 13 and 20, a coil antenna 100A according to a first modification of the present invention comprises a coil 200 and a magnetic core 300A. The coil 200 of the present modification has a structure same as that of the coil 200 of the aforementioned embodiment, and detail explanation thereabout will be omitted.

[0087] Referring to FIG. 20, the magnetic core 300A of the present embodiment is a cast core similar to the aforementioned magnetic core 300. The magnetic core 300A partially forms a magnetic path of the coil 200. The magnetic core 300A is positioned at a region excluding a specific region SR which is positioned just below the coil 200. It is noted that the specific region SR is a region which is hidden below the coil 200 when the coil 200 is viewed from above. The

specific region SR is positioned below a lower end **220** of the coil **200** in the up-down direction. The specific region SR is positioned outside an inner periphery **240** of the coil **200** in a radial direction perpendicular to a winding axis AX. The specific region SR is positioned inside an outer periphery **230** of the coil **200** in the radial direction perpendicular to the winding axis AX. The magnetic core **300A** has a center core **310**.

[0088] Referring to FIGS. **17** and **20**, the center core **310** of the present embodiment has a cylindrical shape extending in the up-down direction. The center core **310** is at least partially positioned inside the coil **200**.

[0089] As shown in FIG. **20**, the center core **310** extends downward in the up-down direction beyond a lower end **220** of the coil **200**. The center core **310** extends upward in the up-down direction beyond an upper end **210** of the coil **200**.

[0090] Referring to FIGS. **10** and **20**, dissimilar to magnetic core **300**, the magnetic core **300A** has no upper portion **320**.

[0091] As shown in FIG. **13**, the coil antenna **100A** comprises a molded resinous member **700A**.

[0092] Referring to FIGS. **14** and **18**, the molded resinous member **700A** covers the coil **200**.

Referring to FIG. **20**, the specific region SR is occupied by the molded resinous member **700A**. In other words, the molded resinous member **700A** is positioned just below the coil **200**. It is noted that an upper end of the center core **310** is not covered with the molded resinous member **700A** as referred to FIG. **15**.

[0093] An antenna device (not shown), which is configured so that the coil antenna **100A** of the present modification is mounted on a circuit board **500**, can also have the same effects as the aforementioned embodiment.

Second Modification

[0094] Referring to FIGS. **22** and **25**, a coil antenna **100B** according to a second modification of the present invention comprises a coil **200** and a magnetic core **300B**. The coil **200** of the present modification has a structure same as that of the coil **200** of the aforementioned embodiment, and detail explanation thereabout will be omitted.

[0095] Referring to FIG. **25**, the magnetic core **300B** of the present embodiment is a cast core similar to the aforementioned magnetic core **300**. The magnetic core **300B** partially forms a magnetic path of the coil **200**. The magnetic core **300B** is positioned at a region excluding a specific region SR which is positioned just below the coil **200**. It is noted that the specific region SR is a region which is hidden below the coil **200** when the coil **200** is viewed from above. The specific region SR is positioned below a lower end **220** of the coil **200** in the up-down direction. The specific region SR is positioned outside an inner periphery **240** of the coil **200** in a radial direction perpendicular to a winding axis AX. The specific region SR is positioned inside an outer periphery **230** of the coil **200** in the radial direction perpendicular to the winding axis AX. The magnetic core **300B** at least has a center core **310**.

[0096] Referring to FIG. **25**, the center core **310** of the present embodiment has a cylindrical shape extending in the up-down direction. The center core **310** is at least partially positioned inside the coil **200**.

[0097] As shown in FIG. **25**, the center core **310** extends downward in the up-down direction beyond a lower end **220** of the coil **200**. The center core **310** extends upward in the up-down direction beyond an upper end **210** of the coil **200**.

[0098] As shown in FIG. **25**, the magnetic core **300B** further has an upper portion **320B**. The upper portion **320B** of the present modification has a structure same as that of the upper portion **320** of the aforementioned embodiment, and detail explanation thereabout will be omitted.

[0099] As shown in FIG. **25**, the magnetic core **300B** further has a side portion **340**.

[0100] As shown in FIG. **25**, the side portion **340** of the present embodiment is positioned outside the coil **200** in a plane perpendicular to the up-down direction. The side portion **340** extends downward in the up-down direction from the upper portion **320B**. This enables a flux of the coil

200, which passes through the upper portion **320B**, to efficiently flow toward a lower part of the coil **200**. The side portion **340** covers the whole of the outer periphery **230** of the coil **200**.

However, the present invention is not limited thereto, but the side portion **340** may cover only a part of the outer periphery **230** of the coil **200**.

[0101] Referring to FIG. **25**, the coil antenna **100B** comprises a molded resinous member (not shown).

[0102] Referring to FIG. **25**, the molded resinous member of the present modification covers the coil **200**. The specific region **SR** is occupied by the molded resinous member. In other words, the molded resinous member is positioned just below the coil **200**.

[0103] An antenna device (not shown), which is configured so that the coil antenna **100B** of the present modification is mounted on a circuit board **500**, can also have the same effects as the aforementioned embodiment.

Second Embodiment

[0104] As shown in FIG. **27**, an antenna device **10C** according to a second embodiment of the present invention comprises a circuit board **500C**, a coil antenna **100** and a resonance circuit. The coil antenna **100** of the present embodiment has a structure same as that of the coil antenna **100** of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0105] As shown in FIG. **27**, the circuit board **500C** of the present embodiment has a conductive portion **510C** with a flat shape. In more detail, the conductive portion **510C** is a ground conductor. Specifically, the circuit board **500C** is configured so that a conductive layer, which functions as the conductive portion **510C**, is laminated on an insulating substrate such as a glass epoxy substrate.

[0106] As shown in FIG. **27**, the conductive portion **510C** is formed with an opening **512** and a slit **514C**. The opening **512** of the present embodiment has a structure same as that of the opening **512** of the first embodiment, and detail explanation thereabout will be omitted. The conductive portion **510C** has a first path which goes around the opening **512** and the slit **514C**. The first path forms a part of the resonance circuit.

[0107] As shown in FIG. **27**, similar to the opening **512**, the slit **514C** of the present embodiment is a part of the circuit board **500C** which has no conductive pattern. The slit **514C** is coupled with the opening **512**. The slit **514C** extends from the opening **512**. The slit **514C** does not communicate with the outside of the conductive portion **510C** in a plane perpendicular to the up-down direction. The slit **514C** has a predetermined portion **PP**. The slit **514C** has a first portion **5142** and a second portion **5144**.

[0108] As shown in FIG. **27**, a width of the slit **514C** is smaller than a diameter of the opening **512**. However, the present invention is not limited thereto, but the width of the slit **514C** may be equal to the diameter of the opening **512**, or the width of the slit **514C** may be greater than the diameter of the opening **512**. It is noted that an increase of the width of the slit **514C** enlarges a current loop, which is formed around the slit **514C**, so that a communication range of the antenna device **10C** is increased.

[0109] As shown in FIG. **27**, the first portion **5142** extends rightward in the right-left direction from the opening **512**.

[0110] As shown in FIG. **27**, the second portion **5144** extends from the first portion **5142** in a direction intersecting with the first portion **5142**. The second portion **5144** extends rearward in the front-rear direction from a right end of the first portion **5142**. The second portion **5144** functions as the predetermined portion **PP**.

[0111] Referring to FIGS. **2** and **27**, dissimilar to the slit **514** of the first embodiment, the slit **514C** does not comprise a third portion **5146**.

[0112] As shown in FIG. **27**, the circuit board **500C** has an edge **520**.

[0113] As shown in FIG. **27**, the edge **520** of the present embodiment defines an outer edge of the

circuit board **500C**. Specifically, the edge **520** defines the outer edge of the circuit board **500C** in a direction perpendicular to the up-down direction. The first portion **5142** extends from the opening **512** toward the edge **520**. The predetermined portion PP extends along the edge **520** in the vicinity of the edge **520**. This configuration enables the circuit board **500C** to have an increased region where electronic components can be arranged on the circuit board **500C**, and the communication range of the antenna device **10C** can be increased by enlarging the current loop, which is formed around the slit **514C**, because this configuration can increase a length of the slit **514C**.

[0114] As shown in FIG. **27**, the antenna device **10C** further has a capacitor **600** which forms a part of the resonant circuit.

[0115] Referring to FIG. **27**, the capacitor **600** forms a current loop path together

[0116] with the first path which goes around the opening **512** and the slit **514C** of the conductive portion **510C**. In other words, the capacitor **600** forms the resonant circuit together with the first path which goes around the opening **512** and the slit **514C** of the conductive portion **510C**. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device **10C**. As shown in FIG. **27**, the capacitor **600** is mounted on the circuit board **500C** at its region between the edge **520** and the slit **514C**. In more detail, the capacitor **600** is mounted on the circuit board **500C** at its region between the edge **520** and the second portion **5144**. However, the mounting location of the capacitor **600** is not limited to that described above, but the capacitor **600** may be mounted at any location on the conductive portion **510C**, provided that the capacitor **600** and the first path, which goes around the opening **512** and the slit **514C** of the conductive portion **510C**, form the current loop path.

Third Embodiment

[0117] As shown in FIG. **28**, an antenna device **10D** according to a third embodiment of the present invention comprises a circuit board **500D**, a coil antenna **100** and a resonance circuit. The coil antenna **100** of the present embodiment has a structure same as that of the coil antenna **100** of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0118] As shown in FIG. **28**, the circuit board **500D** of the present embodiment has a conductive portion **510D** with a flat shape. In more detail, the conductive portion **510D** is a ground conductor. Specifically, the circuit board **500D** is configured so that a conductive layer, which functions as the conductive portion **510D**, is laminated on an insulating substrate such as a glass epoxy substrate.

[0119] As shown in FIG. **28**, the conductive portion **510D** is formed with an opening **512** and a slit **514D**. The conductive portion **510D** has a first path which goes around the opening **512** and the slit **514D**. The first path forms a part of the resonance circuit. The opening **512** of the present embodiment has a structure same as that of the opening **512** of the first embodiment, and detail explanation thereabout will be omitted.

[0120] As shown in FIG. **28**, similar to the opening **512**, the slit **514D** of the present embodiment is a part of the circuit board **500D** which has no conductive pattern. The slit **514D** is coupled with the opening **512**. The slit **514D** extends from the opening **512**. The slit **514D** extends from the opening **512** in the front-rear direction. The slit **514D** does not communicate with the outside of the conductive portion **510D** in a plane perpendicular to the up-down direction.

[0121] As shown in FIG. **28**, a width of the slit **514D** is smaller than a diameter of the opening **512**. However, the present invention is not limited thereto, but the width of the slit **514D** may be equal to the diameter of the opening **512**, or the width of the slit **514D** may be greater than the diameter of the opening **512**. It is noted that an increase of the width of the slit **514D** enlarges a current loop, which is formed around the slit **514D**, so that a communication range of the antenna device **10D** is increased.

[0122] As shown in FIG. **28**, the conductive portion **510D** is further formed with an additional opening **516**.

[0123] As shown in FIG. 28, the additional opening 516 of the present embodiment is a hole piercing the circuit board 500D. In other words, the additional opening 516 pierces the circuit board 500D in the up-down direction. However, the present invention is not limited thereto, but the additional opening 516 may be a part of the circuit board 500D, which has no conductive pattern, similar to the opening 512. The additional opening 516 is positioned forward of the opening 512 in the front-rear direction. In the plane perpendicular to the up-down direction, the additional opening 516 is greater than the opening 512. The conductive portion 510D has a second path which goes around the additional opening 516. The second path forms a part of the resonance circuit. The slit 514D is coupled with both of the opening 512 and the additional opening 516.

[0124] As shown in FIG. 28, the conductive portion 510D is further formed with an additional slit 518.

[0125] As shown in FIG. 28, the additional slit 518 of the present embodiment is a part of the circuit board 500D, which has no conductive pattern, similar to the opening 512. The additional slit 518 is coupled with the opening 512. The additional slit 518 extends from the opening 512. The additional slit 518 extends in the right-left direction. The additional slit 518 communicates with the outside of the conductive portion 510D in the plane perpendicular to the up-down direction. All of the opening 512, the slit 514D, the additional opening 516 and the additional slit 518 are coupled to one another.

[0126] As shown in FIG. 28, a width of the additional slit 518 is smaller than the diameter of the opening 512. However, the present invention is not limited thereto, but the width of the additional slit 518 may be equal to the diameter of the opening 512, or the width of the additional slit 518 may be greater than the diameter of the opening 512.

[0127] As shown in FIG. 28, the circuit board 500D has an edge 520.

[0128] As shown in FIG. 28, the edge 520 defines an outer edge of the circuit board 500. Specifically, the edge 520 defines the outer edge of the circuit board 500D in a direction perpendicular to the up-down direction. The slit 514D does not reach the edge 520. The additional slit 518 extends toward the edge 520. Specifically, the additional slit 518 extends from the opening 512 toward the edge 520. The additional slit 518 reaches the edge 520.

[0129] Referring to FIG. 28, the coil antenna 100 is mounted on the circuit board 500D so that the coil antenna 100 at least overlaps with the opening 512 when the coil antenna 100 is viewed along the up-down direction perpendicular to the circuit board 500D. The coil antenna 100 is mounted on the circuit board 500D so that the coil antenna 100 does not overlap with the additional opening 516 when the coil antenna 100 is viewed along the up-down direction. The coil antenna 100 is surface-mounted on the circuit board 500D.

[0130] As shown in FIG. 28, the antenna device 10D further has capacitors 600D which form a part of the resonant circuit.

[0131] Referring to FIG. 28, the capacitors 600D form a current loop path together with a path which goes around the opening 512, the additional opening 516, the slit 514D and the additional slit 518 of the conductive portion 510D. In other words, the capacitors 600D form the resonant circuit together with the path which goes around the opening 512, the additional opening 516, the slit 514D and the additional slit 518 of the conductive portion 510D. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device 10D. The capacitors 600D are mounted on the circuit board 500D so that each of the capacitors 600D is wired across the additional slit 518. However, the mounting locations of the capacitors 600D are not limited to those described above, but the capacitors 600D may be mounted at any locations on the conductive portion 510D, provided that the capacitors 600D and the path, which goes around the opening 512, the additional opening 516, the slit 514D and the additional slit 518 of the conductive portion 510D, form the current loop path.

Fourth Embodiment

[0132] As shown in FIG. 29, an antenna device 10E according to a fourth embodiment of the

present invention comprises a circuit board **500E**, a coil antenna **100** and a resonance circuit. The coil antenna **100** of the present embodiment has a structure same as that of the coil antenna **100** of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0133] As shown in FIG. **29**, the circuit board **500E** of the present embodiment has a conductive portion **510E** with a flat shape. In more detail, the conductive portion **510E** is a ground conductor. Specifically, the circuit board **500E** is configured so that a conductive layer, which functions as the conductive portion **510E**, is laminated on an insulating substrate such as a glass epoxy substrate.

[0134] As shown in FIG. **29**, the conductive portion **510E** is formed with an opening **512** and a slit **514E**. The conductive portion **510E** has a first path which goes around the opening **512** and the slit **514E**. The first path forms a part of the resonance circuit. The opening **512** of the present embodiment has a structure same as that of the opening **512** of the first embodiment, and detail explanation thereabout will be omitted.

[0135] As shown in FIG. **29**, similar to the opening **512**, the slit **514E** of the present embodiment is a part of the circuit board **500E** which has no conductive pattern. The slit **514E** is coupled with the opening **512**. The slit **514E** extends leftward in the right-left direction from the opening **512**, and then extends rearward in the front-rear direction. The slit **514E** extends from the opening **512**. The slit **514E** does not communicate with the outside of the conductive portion **510E** in a plane perpendicular to the up-down direction.

[0136] As shown in FIG. **29**, a width of the slit **514E** is smaller than a diameter of the opening **512**. However, the present invention is not limited thereto, but the width of the slit **514E** may be equal to the diameter of the opening **512**, or the width of the slit **514E** may be greater than the diameter of the opening **512**. It is noted that an increase of the width of the slit **514E** enlarges a current loop, which is formed around the slit **514E**, so that a communication range of the antenna device **10E** is increased.

[0137] As shown in FIG. **29**, the conductive portion **510E** is further formed with an additional opening **516**.

[0138] As shown in FIG. **29**, the additional opening **516** of the present embodiment is a hole piercing the circuit board **500E**. In other words, the additional opening **516** pierces the circuit board **500E** in the up-down direction. However, the present invention is not limited thereto, but the additional opening **516** may be a part of the circuit board **500E**, which has no conductive pattern, similar to the opening **512**. The additional opening **516** is positioned rearward of the opening **512** in the front-rear direction. In the plane perpendicular to the up-down direction, the additional opening **516** is greater than the opening **512**. The conductive portion **510E** has a second path which goes around the additional opening **516**. The second path forms a part of the resonance circuit. The slit **514E** is coupled with both of the opening **512** and the additional opening **516**.

[0139] As shown in FIG. **29**, the conductive portion **510E** is further formed with an additional slit **518**.

[0140] As shown in FIG. **29**, the additional slit **518** of the present embodiment is a part of the circuit board **500E**, which has no conductive pattern, similar to the opening **512**. The additional slit **518** is coupled with the additional opening **516**. The additional slit **518** extends from the additional opening **516**. The additional slit **518** extends in the right-left direction. The additional slit **518** communicates with the outside of the conductive portion **510E** in the plane perpendicular to the up-down direction. All of the opening **512**, the slit **514E**, the additional opening **516** and the additional slit **518** are coupled to one another.

[0141] As shown in FIG. **29**, the circuit board **500E** has an edge **520**.

[0142] As shown in FIG. **29**, the edge **520** defines an outer edge of the circuit board **500E**.

Specifically, the edge **520** defines the outer edge of the circuit board **500E** in a direction perpendicular to the up-down direction. The slit **514E** does not reach the edge **520**. The additional

slit **518** extends toward the edge **520**. Specifically, the additional slit **518** extends from the additional opening **516** toward the edge **520**. The additional slit **518** reaches the edge **520**.

[0143] Referring to FIG. **29**, the coil antenna **100** is mounted on the circuit board **500E** so that the coil antenna **100** at least overlaps with the opening **512** when the coil antenna **100** is viewed along the up-down direction perpendicular to the circuit board **500E**. The coil antenna **100** is mounted on the circuit board **500E** so that the coil antenna **100** does not overlap with the additional opening **516** when the coil antenna **100** is viewed along the up-down direction. The coil antenna **100** is surface-mounted on the circuit board **500E**.

[0144] As shown in FIG. **29**, the antenna device **10E** further has capacitors **600E** which form a part of the resonant circuit.

[0145] Referring to FIG. **29**, the capacitors **600E** form a current loop path together with a path which goes around the opening **512**, the additional opening **516**, the slit **514E** and the additional slit **518** of the conductive portion **510E**. In other words, the capacitors **600E** form the resonant circuit together with the path which goes around the opening **512**, the additional opening **516**, the slit **514E** and the additional slit **518** of the conductive portion **510E**. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device **10E**. The capacitors **600E** are mounted on the circuit board **500E** so that each of the capacitors **600E** is wired across the additional slit **518**. However, the mounting locations of the capacitors **600E** are not limited to those described above, but the capacitors **600E** may be mounted at any locations on the conductive portion **510E**, provided that the capacitors **600E** and the path, which goes around the opening **512**, the additional opening **516**, the slit **514E** and the additional slit **518** of the conductive portion **510E**, form the current loop path.

Fifth Embodiment

[0146] As shown in FIG. **30**, an antenna device **10F** according to a fifth embodiment of the present invention comprises a circuit board **500F**, a coil antenna **100** and a resonance circuit. The coil antenna **100** of the present embodiment has a structure same as that of the coil antenna **100** of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0147] As shown in FIG. **30**, the circuit board **500F** of the present embodiment has a conductive portion **510F** with a flat shape. In more detail, the conductive portion **510F** is a ground conductor. Specifically, the circuit board **500F** is configured so that a conductive layer, which functions as the conductive portion **510F**, is laminated on an insulating substrate such as a glass epoxy substrate.

[0148] As shown in FIG. **30**, the conductive portion **510F** is formed with an opening **512** and a slit **514F**. The conductive portion **510F** has a first path which goes around the opening **512** and the slit **514F**. The first path forms a part of the resonance circuit. The opening **512** of the present embodiment has a structure same as that of the opening **512** of the first embodiment, and detail explanation thereabout will be omitted.

[0149] As shown in FIG. **30**, similar to the opening **512**, the slit **514F** of the present embodiment is a part of the circuit board **500F** which has no conductive pattern. The slit **514F** is coupled with the opening **512**. The slit **514F** extends from the opening **512**.

[0150] As shown in FIG. **30**, a width of the slit **514F** is smaller than a diameter of the opening **512**. However, the present invention is not limited thereto, but the width of the slit **514F** may be equal to the diameter of the opening **512**, or the width of the slit **514F** may be greater than the diameter of the opening **512**. It is noted that an increase of the width of the slit **514F** enlarges a current loop, which is formed around the slit **514F**, so that a communication range of the antenna device **10F** is increased.

[0151] As shown in FIG. **30**, the conductive portion **510F** is further formed with an additional opening **516**.

[0152] As shown in FIG. **30**, the additional opening **516** of the present embodiment is a hole

piercing the circuit board **500F**. In other words, the additional opening **516** pierces the circuit board **500F** in the up-down direction. However, the present invention is not limited thereto, but the additional opening **516** may be a part of the circuit board **500F**, which has no conductive pattern, similar to the opening **512**. The additional opening **516** is positioned rearward of the opening **512** in the front-rear direction. In a plane perpendicular to the up-down direction, the additional opening **516** is greater than the opening **512**. The conductive portion **510F** has a second path which goes around the additional opening **516**. The second path forms a part of the resonance circuit. The slit **514F** is coupled with both of the opening **512** and the additional opening **516**.

[0153] As shown in FIG. **30**, the conductive portion **510F** is further formed with an additional slit **518**.

[0154] As shown in FIG. **30**, the additional slit **518** of the present embodiment is a part of the circuit board **500F**, which has no conductive pattern, similar to the opening **512**. The additional slit **518** is coupled with the slit **514F**. The additional slit **518** extends in the right-left direction from the slit **514F**. The additional slit **518** communicates with the outside of the conductive portion **510F** in the plane perpendicular to the up-down direction. In the plane perpendicular to the up-down direction, the slit **514F** communicates with the outside of the conductive portion **510F** via the additional slit **518**. All of the opening **512**, the slit **514F**, the additional opening **516** and the additional slit **518** are coupled to one another.

[0155] As shown in FIG. **30**, a width of the additional slit **518** is equal to the width of the slit **514F**. However, the present invention is not limited thereto, but the width of the additional slit **518** may be smaller than the width of the slit **514F** or may be greater than the width of the slit **514F**.

[0156] As shown in FIG. **30**, the circuit board **500F** has an edge **520**.

[0157] As shown in FIG. **30**, the edge **520** defines an outer edge of the circuit board **500F**.

Specifically, the edge **520** defines the outer edge of the circuit board **500F** in a direction perpendicular to the up-down direction. The additional slit **518** extends toward the edge **520**.

Specifically, the additional slit **518** extends from the slit **514F** toward the edge **520**. The additional slit **518** reaches the edge **520**.

[0158] Referring to FIG. **30**, the coil antenna **100** is mounted on the circuit board **500F** so that the coil antenna **100** at least overlaps with the opening **512** when the coil antenna **100** is viewed along the up-down direction perpendicular to the circuit board **500F**. The coil antenna **100** is mounted on the circuit board **500F** so that the coil antenna **100** does not overlap with the additional opening **516** when the coil antenna **100** is viewed along the up-down direction. The coil antenna **100** is surface-mounted on the circuit board **500F**.

[0159] As shown in FIG. **30**, the antenna device **10F** further has capacitors **600F** which form a part of the resonant circuit.

[0160] Referring to FIG. **30**, the capacitors **600F** form a current loop path together with a path which goes around the opening **512**, the additional opening **516**, the slit **514F** and the additional slit **518** of the conductive portion **510F**. In other words, the capacitors **600F** form the resonant circuit together with the path which goes around the opening **512**, the additional opening **516**, the slit **514F** and the additional slit **518** of the conductive portion **510F**. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device **10F**. The capacitors **600F** are mounted on the circuit board **500F** so that each of the capacitors **600F** is wired across the additional slit **518**. However, the mounting locations of the capacitors **600F** are not limited to those described above, but the capacitors **600F** may be mounted at any locations on the conductive portion **510F**, provided that the capacitors **600F** and the path, which goes around the opening **512**, the additional opening **516**, the slit **514F** and the additional slit **518** of the conductive portion **510F**, form the current loop path.

[0161] Although the specific explanation about the present invention is made above referring to the embodiments, the present invention is not limited thereto and is susceptible to various modifications and alternative forms. In addition, the above embodiments and variations may also

be combined.

[0162] Although each of the antenna devices **10**, **10C** of the aforementioned first and second embodiments is not provided with a portion equivalent to the additional opening **516** of each of the antenna devices **10D**, **10E**, **10F** of the third, fourth and fifth embodiments, the present invention is not limited thereto. Specifically, the antenna device **10**, **10C** may also be provided with an additional opening **516** which forms a part of the slit **514**, **514C**.

[0163] The present application is based on a Japanese patent applications of JP2022-074955 filed before the Japan Patent Office on Apr. 28, 2022 and JP2022-118580 filed before the Japan Patent Office on Jul. 26, 2022, the contents of which are incorporated herein by reference.

[0164] While there has been described what is believed to be the preferred embodiment of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such embodiments that fall within the true scope of the invention.

REFERENCE SIGNS LIST

[0165] **10**, **10C**, **10D**, **10E**, **10F** antenna device [0166] **100**, **100A**, **100B** coil antenna [0167] **200** coil [0168] **210** upper end [0169] **220** lower end [0170] **230** outer periphery [0171] **240** inner periphery [0172] **300**, **300A**, **300B** magnetic core [0173] **310** center core [0174] **320**, **320B** upper portion [0175] **340** side portion [0176] **400** terminal [0177] **500**, **500C**, **500D**, **500E**, **500F** circuit board [0178] **510**, **510C**, **510D**, **510E**, **510F** conductive portion [0179] **512** opening [0180] **514**, **514C**, **514D**, **514E**, **514F** slit [0181] **5142** first portion [0182] **5144** second portion [0183] **5146** third portion [0184] **516** additional opening [0185] **518** additional slit [0186] **520** edge [0187] **600**, **600D**, **600E**, **600F** capacitor [0188] **700**, **700A** molded resinous member [0189] **AX** winding axis [0190] **PP** predetermined portion [0191] **SR** specific region

Claims

1. An antenna device comprising a circuit board, a coil antenna and a resonance circuit, wherein: the circuit board has a conductive portion with a flat shape; the conductive portion is formed with an opening and a slit; the conductive portion has a first path which goes around the opening and the slit; the first path forms a part of the resonance circuit; the slit is coupled with the opening; the coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along an up-down direction perpendicular to the circuit board; the coil antenna comprises a coil and a magnetic core; the coil has a winding axis extending in the up-down direction; the magnetic core partially forms a magnetic path of the coil; the magnetic core is positioned at a region excluding a specific region which is positioned just below the coil; the magnetic core at least has a center core; and the center core is at least partially positioned inside the coil.

2. The antenna device as recited in claim 1, wherein: the circuit board has an edge; and the slit extends toward the edge.

3. The antenna device as recited in claim 1, wherein: the conductive portion is further formed with an additional opening; the conductive portion has a second path which goes around the additional opening; the second path forms a part of the resonance circuit; the slit is coupled with both of the opening and the additional opening; and the coil antenna is mounted on the circuit board so that the coil antenna does not overlap with the additional opening when the coil antenna is viewed along the up-down direction.

4. The antenna device as recited in claim 3, wherein: the circuit board has an edge; the conductive portion is further formed with an additional slit; the additional slit extends toward the edge; the antenna device further has a capacitor which forms a part of the resonant circuit; and the capacitor is mounted on the circuit board so that the capacitor is wired across the additional slit.

5. The antenna device as recited in claim 4, wherein all of the opening, the slit, the additional

opening and the additional slit are coupled to one another.

6. The antenna device as recited in claim 1, wherein: the coil has a lower end in the up-down direction; and the center core extends downward in the up-down direction beyond the lower end.

7. The antenna device as recited in claim 1, wherein: the coil has an upper end in the up-down direction; and the center core extends upward in the up-down direction beyond the upper end.

8. The antenna device as recited in claim 7, wherein: the coil has an outer periphery; the magnetic core further has an upper portion; the upper portion extends from the center core toward the outer periphery; and the upper portion overlaps with the coil when the upper portion is viewed in the up-down direction.

9. The antenna device as recited in claim 8, wherein: the magnetic core further has a side portion; the side portion is positioned outside the coil in a plane perpendicular to the up-down direction; and the side portion extends downward in the up-down direction from the upper portion.

10. The antenna device as recited in claim 1, wherein: the circuit board has an edge; the slit has a predetermined portion; and the predetermined portion extends along the edge in the vicinity of the edge.

11. The antenna device as recited in claim 10, wherein: the antenna device further has a capacitor which forms a part of the resonant circuit; and the capacitor is mounted on the circuit board at its region between the edge and the slit.

12. The antenna device as recited in claim 10, wherein: the slit has a first portion and a second portion; the first portion extends from the opening toward the edge; the second portion extends from the first portion in a direction intersecting with the first portion; and the second portion functions as the predetermined portion.

13. The antenna device as recited in claim 12, wherein: the slit further comprises a third portion; and the third portion extends from the second portion in a direction intersecting with the second portion.

14. The antenna device as recited in claim 1, wherein: the coil has an inner periphery; and $S1$ and $S2$ satisfy $1 \leq S2/S1 \leq 2$, where $S1$ is a cross-sectional area of the inner periphery of the coil, and $S2$ is an area of the opening.
